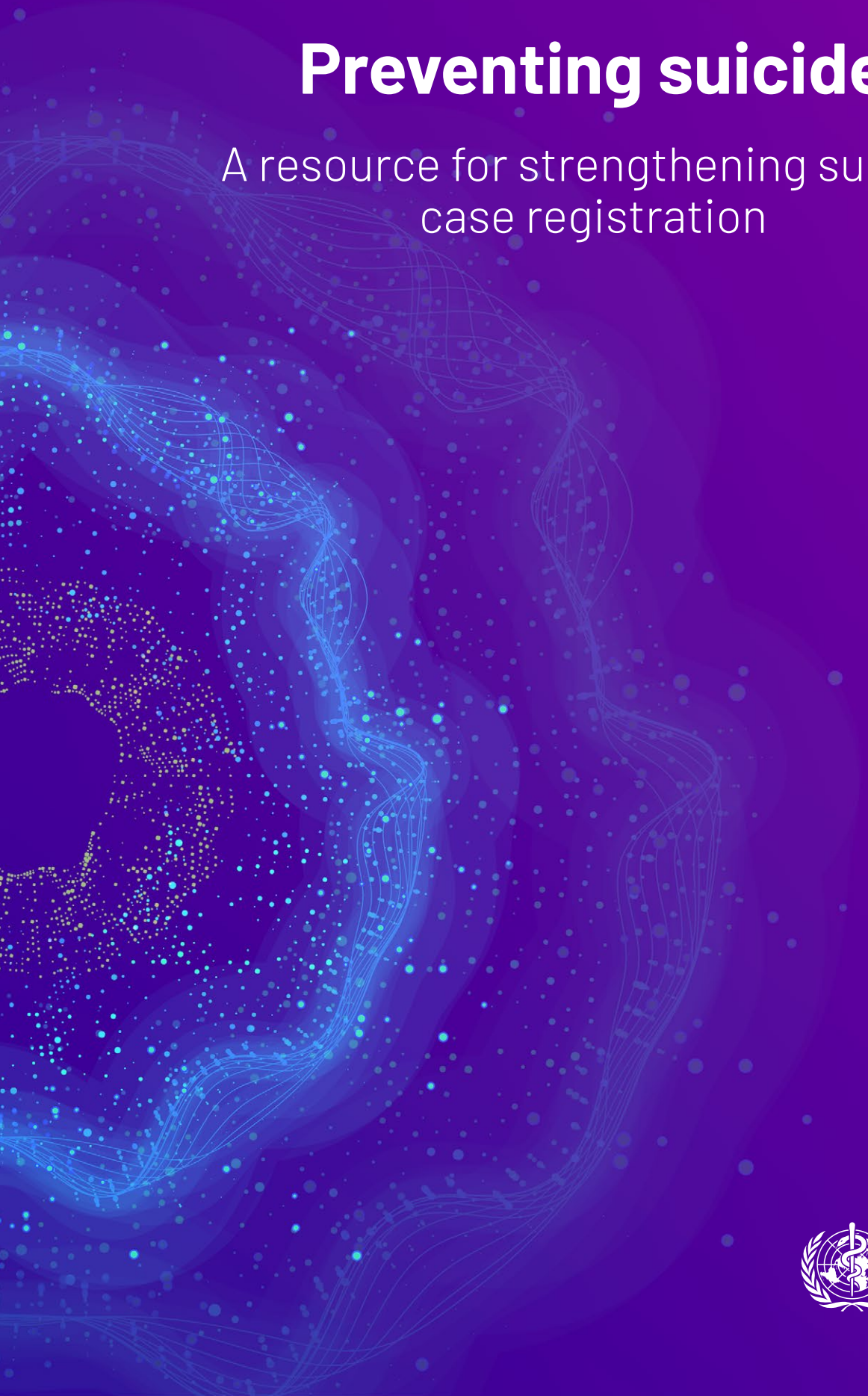
Preventing suicide

A resource for strengthening suicide case registration



World Health Organization

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Foreword

Suicide remains a serious global public health concern that demands urgent attention and requires a whole-of-government and whole-of-society approach. Evidence shows that suicide prevention is possible but that it necessitates a comprehensive range of actions. These include creating supportive environments for children and young people, limiting access to the means of suicide, promoting responsible media reporting, and ensuring timely and accurate identification and management of suicidal thoughts, mental health conditions and other important risk factors.

Critical to the prevention of suicide is the availability of high-quality, comprehensive data, ideally collected through civil registration and vital statistics (CRVS) systems. Such data help to highlight the scale of the issue and provide essential information – such as age, sex and methods of suicide – that inform the development of timely, evidence-based prevention strategies. WHO's *LIVE LIFE: an implementation guide for suicide prevention in countries* (WHO, 2021) further underscores the critical role of accurate suicide data in prevention efforts.

Given the particular challenges associated with registering deaths by suicide, this guidance aims to strengthen the death certification and coding for suicides. The guidance is primarily intended for professionals involved in certifying deaths and for mortality coders, but it may also be useful for other professionals involved in suicide death registration. The document outlines key approaches to the registration of suicide cases and provides examples of how to certify and code suicide deaths in order to ensure good-quality data.

This resource is a revision of Preventing suicide: a resource for suicide case registration, published in 2011, and builds on previous work related to medical certification of the cause of death and surveillance of self-harm in both health-care and community settings. The resource represents a critical link in a broad and collaborative effort involving diverse stakeholders – governments, health professionals, medico-legal officers, educators, social agencies, legislators, law enforcers, statisticians, social communicators, media professionals, religious leaders, people with lived experience of suicide or self-harm, and communities – all committed to advancing suicide prevention.

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Reference

WHO, 2021. LIVE LIFE: an implementation guide for suicide prevention in countries. Geneva: World Health Organization; 2021 (https://www.who.int/publications/i/item/9789240026629, accessed 9 April 2025).

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WHO is particularly indebted to Professor Kairi Kolves (Australian Institute for Suicide Research and Prevention, Griffith University, Australia) who undertook the update of the 2011 version of the resource to arrive at this current version. WHO is also grateful to Professor Stephen Platt (University of Edinburgh, Edinburgh, United Kingdom of Great Britain and Northern Ireland) for contributing to Annex 1. Further revisions were made by Dr Alexandra Fleischmann (WHO) and Dr Piumee Bandara (WHO).

The text was subsequently reviewed by the following WHO specialists and other international experts.

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This resource is being disseminated widely in the hope that it will be translated and adapted to local contexts, which is a prerequisite for its effectiveness. WHO welcomes comments and requests for permission to translate and adapt the resource.

A single chair reflecting on a water surface on a stormy day. © Freepik.

1. Introduction

Surveillance and monitoring together form a core pillar for the prevention of suicide *(1)*. Timely, accurate and complete data on causes of death provide essential information for quantifying the size of the problem and identifying vulnerable populations in terms of age, sex and other meaningful categories. This information is fundamental to the development of suicide prevention strategies. Surveillance data can also provide information on trends in suicide over time and are therefore a valuable tool in priority-setting, public health practice, research, and evaluation of interventions and policy changes *(1)*. Importantly, high-quality suicide mortality data are crucial in monitoring progress towards the United Nations' Sustainable Development Goals (SDGs) target 3.4 and indicator 3.4.2 to reduce suicide mortality by one third by 2030 *(2)*.

1.1 The purpose and development of this resource booklet

The quality of suicide mortality statistics depends on the accuracy of certifying deaths due to suicide and coding the underlying cause of death. The purpose of the present resource is to strengthen the certification and coding of deaths due to suicide. It is primarily designed to support: 1) individuals who are completing the medical certificate of the cause of death; and 2) mortality coders who are involved in coding deaths due to suicide. This resource booklet may also be of interest to other professionals involved in the process of investigating and certifying deaths due to suicide, including police officers, forensic doctors, coroners, physician assistants, nurse practitioners and statisticians. This resource outlines death registration practices specific to suicide. It covers how deaths due to suicide are confirmed and certified, how they are coded using ICD-10 and ICD-11, approaches to ensuring data quality, and procedures for analysing and disseminating suicide data.

Note that this resource should be read as subject to national laws and practices which take precedence in the case of contradiction or doubt. As indicated in WHO's *Recommendations for conducting an external inspection of a body and filling in the Medical Certificate of Cause of Death*, "it is essential that the certifying physician complies with national law and the guidance provided by professional authorities in all respects, in order to avoid errors which might invalidate medical certification of cause of death or expose them to criminal or civil liability" *(3, p2)*.

This resource is a revision of Preventing suicide: a resource for suicide case registration, published in 2011. The revision was led by Professor Kairi Kolves (Australian Institute for Suicide Research and Prevention, Griffith University, Australia), who sought to strengthen suicide case registration with the same dedication as her predecessor at the Institute, Professor Diego De Leo, who led the drafting of the original publication. The revised version is more comprehensive and aligned with current practices, including inputs to Annex 1 from Professor Stephen Platt (University of Edinburgh, Edinburgh, United Kingdom of Great Britain and Northern Ireland). Once a preliminary version was agreed, the draft was circulated for international and WHO review. A full list of reviewers and their affiliations is provided in the Acknowledgements. Following the first round of review, feedback was incorporated by WHO and a revised draft was shared in two rounds with the WHO Mortality Reference Group, whose feedback was also integrated to arrive at the final version.

International reviewers submitted to WHO a declaration of interests, disclosing potential conflicts of interest that might affect, or might reasonably be perceived to affect, their objectivity and independence in relation to the subject matter of this resource. WHO reviewed each of the declarations and concluded that none could give rise to a potential or reasonably perceived conflict of interest related to the subject matter covered by the resource.

1.2 Challenges in determining and registering deaths due to suicide

A well-functioning civil registration and vital statistics (CRVS) system is the most reliable way to monitor causes of death, including suicide, as articulated in the *WHO civil registration and vital statistics strategic implementation plan 2021–2025 (4)*. A civil registration system functions to register all vital events, including deaths and their causes, and to report them to the vital statistics system for analysis and dissemination. When combined with population data, vital statistics provide crucial public health information that is needed to understand emerging health issues and the burden of health conditions over time *(4)*. However, the overwhelming majority of low- and middle-income countries (LMICs) do not have reliable mortality statistics.[1] Among the 86 countries that had good-quality[2] mortality data for the year 2021, 37 were LMICs – which accounted for only 16% of all estimated suicides in LMICs. WHO has formally recognized the need to strengthen CRVS systems globally *(4)*. Furthermore, the *WHO Recommendations for conducting an external inspection of a body and filling in the Medical Certificate of Cause of Death (3)* have been developed to enhance accurate certification of cause of death, with a focus on unsuspicious, natural deaths (i.e. deaths due to disease).

In the case of unnatural or suspicious deaths, such as a death due to suicide, multiple authorities (e.g. medical doctors, police, coroners, forensic doctors) and systems across government are involved. If there are no strong links between agencies – including data-sharing agreements and information flows – registration processes may become fragmented and complex, resulting in possible misclassification, duplication and loss of data *(5)*. Furthermore, because of challenges in determining intent and distinguishing the manner of death from accident or homicide, a suspected death due to suicide often requires significant time for a medico-legal investigation to be completed.

However, such an investigation may be at odds with cultural or religious obligations (e.g. if the family requires burial/cremation). In addition, the individual completing a cause-of-death form (i.e. the certifier) may not have complete and accurate information about the circumstances that led to the death, especially if the certifier does not have authority to visit the scene of the death or to receive information from the agency (e.g. police) responsible for examining the scene of death. In such circumstances, the certifier may indicate the manner of death as "undetermined". Moreover, inadequate training and infrastructure for accurate certification of a cause of death and subsequent coding of data by mortality coders may also hinder data quality. As a result, data on suicide may be only partially available (e.g. the cause of death may be recorded but a detailed description of the method may be missing). This lack of data severely limits the capacity to analyse statistical trends in suicide and to respond accordingly.

Determining the manner of death as suicide is additionally complicated in some settings due to the criminalization of suicidal behaviour,[3] by insurance policies that may not cover deaths by suicide, or by other sociocultural and religious factors which contribute to stigma and beliefs around suicide. These and other related factors may influence the practices of the certifying doctor or coroner to disguise the death as an accident or to attribute it to natural causes rather than suicide, resulting in under-reporting of suicide. In addition, these factors may also contribute to variations in the ratios of under-reporting across age, sex and ethnicity. For instance, suicides are more likely to be under-reported among women and young people of African descent according to evidence from some high-income countries *(7, 8)*.

[1] See link in Annex 6 to WHO webpage on Suicide data.

[2] "Good-quality" refers to countries with a quality rating of 1 or 2 according to WHO's classification, as follows: 1 = multiple years of national death registration data with high completeness and quality of cause-of-death assignment are available; 2 = multiple years of moderate-quality death registration data are available, data have low completeness and/or issues with cause-of-death assignment which possibly affect estimation of deaths by cause and time trends.

[3] Decriminalization of suicidal behaviour represents one critical step that governments can take in their efforts to destigmatize and prevent suicide. WHO has developed a Policy brief on the health aspects of decriminalization of suicide and suicide attempts to support policy-makers and legislators in decriminalizing suicide and suicide attempts *(6)*.

2. Methods of registering deaths due to suicide

The registration of deaths due to suicide is often a complex, multilevel process that involves input from multiple experts and officials situated within several sectors and agencies. Depending on the jurisdiction, the confirmation, certification and registration of deaths due to suicide may be the exclusive responsibility of the medical system, the police and legal system, or often a combination of both *(9, 10)*. Typically, when a death is determined to be unnatural – i.e. caused in a manner such as suicide, accident or homicide – the legal frameworks of many countries require that the death is examined by the medico-legal death investigation system *(11)*. Regardless of the type of medico-legal death investigation system to which suspicious and unnatural deaths are referred, the body should be examined by a medical doctor and certified, ideally using WHO's international form for medical certification of cause of death (MCCD),[4] based on clinical judgement and following thorough investigation. In situations where resources or geographical access are limited, or where medico-legal death investigation and civil registration systems are weakly linked, other key informants (e.g. family members of the deceased, or individuals from the community) may be interviewed to provide data on deaths due to suicide occurring in the community *(9)*. Importantly, multisectoral collaboration and effective integration of data sources (e.g. hospital, police, community key informants) are needed to ensure that all deaths due to suicide are reported to a centralized national information system – ideally via the national CRVS system, for analysis and dissemination.

[4] See link in Annex 6 to WHO webpage on Reporting cause of death.

Milky fog and bright sun curtains. Luminous morning. © Freepik.

2.1 Civil registration and vital statistics system (CRVS)

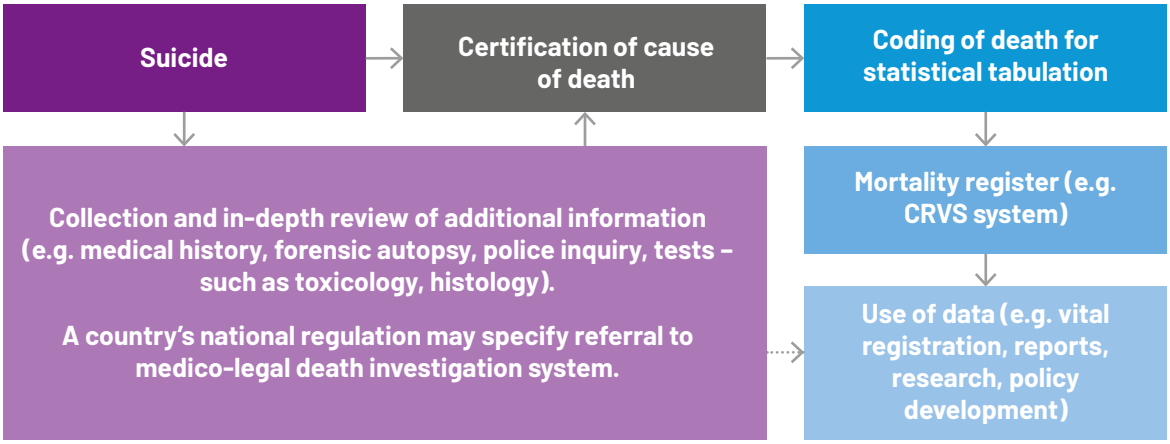
A well-functioning national system of civil registration and vital statistics provides the most reliable source of continuous data on a country's births and all causes of death. CRVS systems rely on medical certification of cause of death which follows the global standard maintained by WHO and the International Classification of Diseases (ICD), with ICD-11 being the latest revision. A simplified workflow of the process of registering deaths due to suicide within a CRVS system can be seen in Fig. 1.

As shown in Fig. 1, the collection of additional information on the death (e.g. by police inquiry and forensic autopsy) or referral to medico-legal death investigation aid in the determination of the cause and manner of death for certification. The extent to which the police and legal system are involved can vary depending on a country's legal framework. In some countries, suicide case registration is primarily the responsibility of the police or coronial system. When a death due to suicide is suspected, the police are informed and the case may be referred to the coroner (i.e. a legal official who may also be a medical doctor).

Additionally, the legal framework in countries may define the deaths that require an autopsy or provide the forensic doctor with authority to determine, or assist in the determination of, the cause and manner of death *(5)*.

Regardless of the process of registration, information about the final cause and manner of death should be submitted to a national statistical office or CRVS system where coders translate the medical information into ICD codes and select the underlying cause of death by applying the ICD rules and principles. In the absence of a well-functioning CRVS system or centralized registry, official statistics on suicide may be stored and managed exclusively by the health sector (e.g. Ministry of Health) or legal system. Further information on how to establish and strengthen a CRVS system can be found in the WHO toolkit on *Improving the quality and use of birth, death and cause-of-death information (12)* and the WHO resource kit on *Strengthening civil registration and vital statistics for births, deaths and causes of death (13)*.

Fig. 1. *Simplified overview of the process for registering and disseminating data on suicide*



2.2 Sample or local surveillance system

National coverage of surveillance may not be feasible in countries where financial resources, infrastructure or appropriately trained staff are inadequate. Geographical remoteness may also have an impact on the ability to capture data on suicide across the whole country. If this is the case, a reporting system specifically for suicide (e.g. a suicide case registry) can be implemented in a sample area (or areas), such as a region or province where quality can be better assured *(1)*. Sample areas should be representative (or as close as possible to representative) of the population in the country as a whole in terms of key characteristics, including sex, age, socioeconomic status, and rural and urban population distribution. Other characteristics of the population which could affect reporting of suicides, such as socio-cultural factors, should also be considered. Coverage can then be expanded once practices are routine, feasible and high-quality, and where information from the system has begun to demonstrate its utility and importance to policy- and decision-makers and other potential funding sources *(1)*. For examples of suicide registry systems worldwide, see *(14)* and *(15)*.

2.3 Verbal autopsy and community key informants

In some jurisdictions, the availability of medical doctors to certify deaths and complete forensic autopsies may be limited. Local traditions and/or religious practices (e.g. a requirement for cremation or burial within 24 hours) may also greatly reduce the time available for examining the body and certifying the death *(16)*. Therefore, where medical certification is not possible, verbal autopsy may also be used to determine the cause of death from the next of kin of the deceased. Verbal autopsy involves interviewing the primary companion or caregiver (e.g. an adult family member) of the deceased using a structured questionnaire to obtain retrospectively key information to be used in identifying a probable cause and manner of death. WHO has developed a verbal autopsy instrument for routine use, with accompanying training materials to assist in standardization of data collection.[5] The WHO instrument includes a verbal autopsy cause of death list that can be translated into ICD-10 or ICD-11 codes in order to complement and be integrated into a CRVS system. In addition to verbal autopsy, community key informants (e.g. community leaders, primary health workers, religious leaders) may be able to provide information on cases of suicide occurring in their community, in the absence of formal surveillance systems.

Community key informant oral and/or written statements should be corroborated with other sources of data (e.g. police and hospital information where available) to ensure accuracy and assist in comprehensively capturing deaths due to suicide *(9)*. Further detail on this approach and on the training materials can be found in WHO's *Training manual for surveillance of suicide and self-harm in communities via key informants (9)*. While information from third-party informants may supplement other sources of data, it is important to acknowledge that stigma and the criminalization of suicide and suicide attempts in some countries may deter the reporting of cases. This form of community surveillance is not a substitute for a well-functioning CRVS but can be considered an adjunct to capture deaths due to suicide.

[5] See link in Annex 6 to WHO's Verbal autopsy standards: ascertaining and attributing causes of death tool.

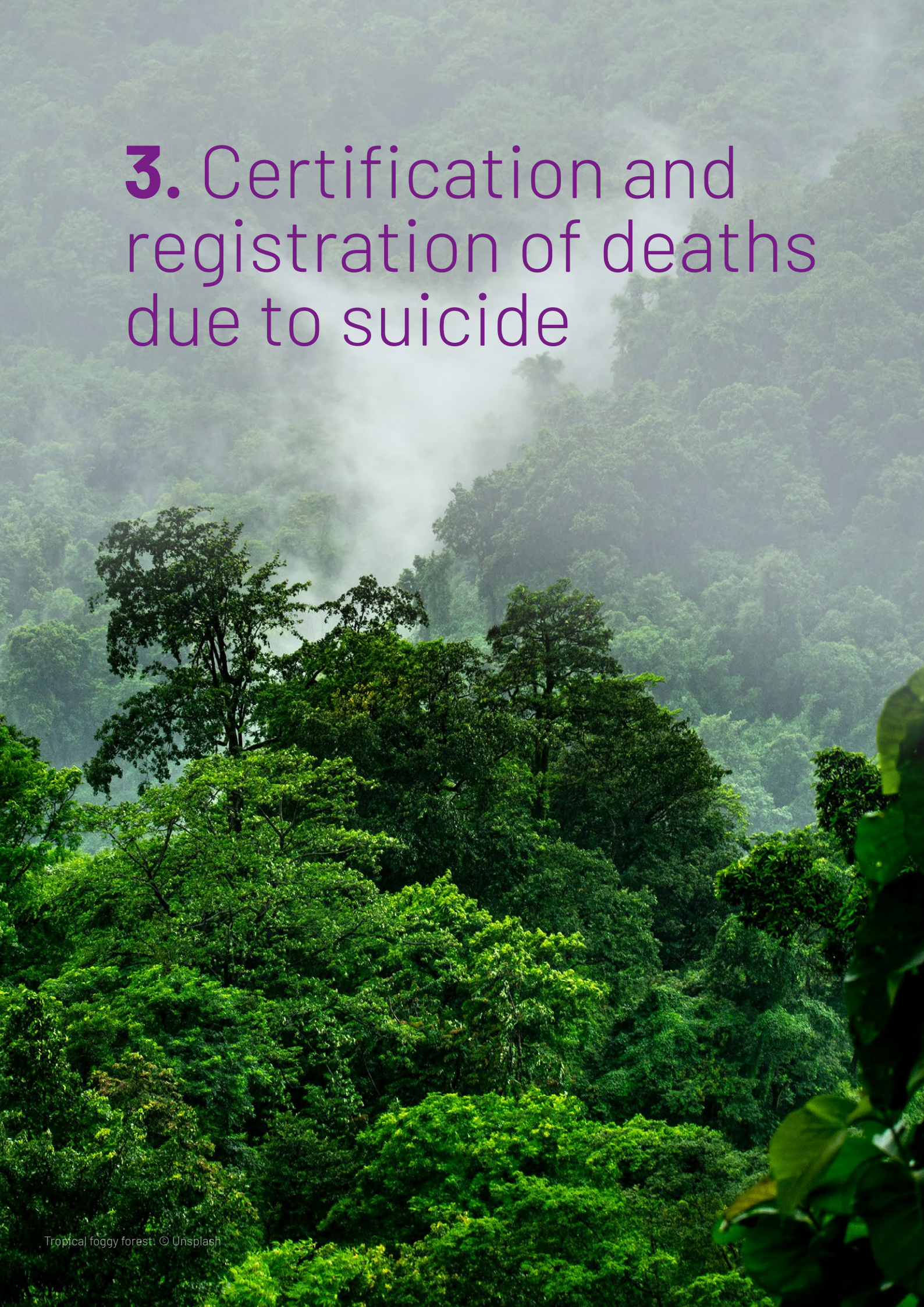
2.4 Household surveys with a mortality module

For countries that do not have a functional CRVS system or routine system of surveillance, household surveys with mortality modules may also be a useful method for capturing the deaths due to suicide in the general population at a given point or period of time. However, important consideration should be given to ensuring that the survey sample is representative of the general population. The WHO STEPS survey is a standardized method for collecting, analysing and disseminating data on key noncommunicable disease risk factors in countries. The STEPS survey includes an optional module on "Mental health/suicide" with questions related to suicidal thoughts, suicide attempts and suicide death by a close family member. Further detail on the STEPS instrument, planning, sampling and reporting can be found on the WHO STEPS survey website.[6]

If households identified deaths, verbal autopsies could be linked to a household survey. As in the case of retrospective verbal autopsy and key informant reporting, third-party information obtained from household surveys can be subject to bias and possible underreporting. Furthermore, such measures are often resource- and time-intensive and are limited by the reference period of measurement (e.g. once or periodically every 5 or 10 years)*(9)*.

[6] For details of the STEPwise approach to NCD risk factor surveillance (STEPS), see link in Annex 6.

3. Certification and registration of deaths due to suicide



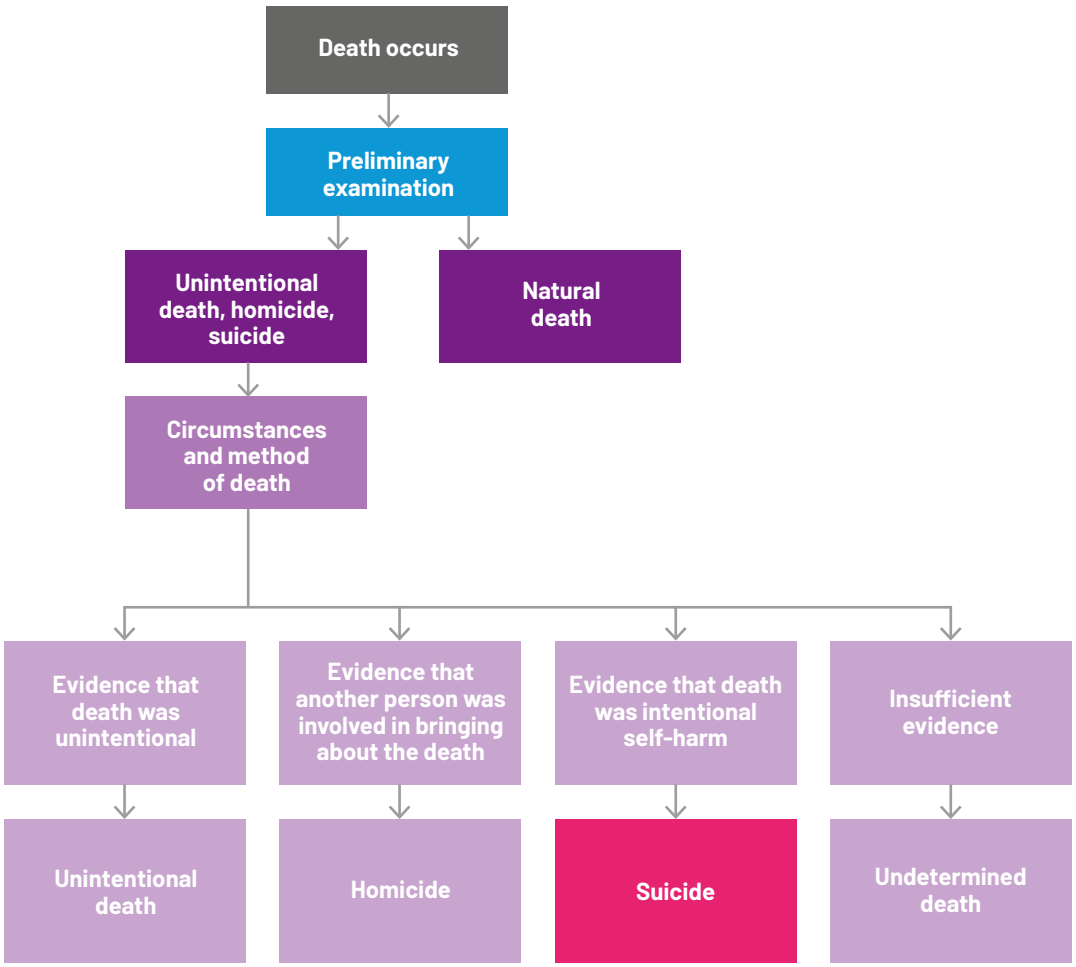
Tropical foggy forest. © Unsplash

3.1 Key information to assist in determining a death due to suicide

According to the ICD, death by suicide is categorized as "intentional self-harm" and is further subclassified according to the method of intentional self-harm. Given that suicide is fatal, establishing intention to die (which can fluctuate, be transient or ambivalent) can be challenging *(17)*. The certifier and/or coder may not have all the contextual information available to assess intent and to distinguish between cases of accident (unintentional death according to ICD-11 terminology), homicide and suicide (Fig. 2).

Suicide by self-poisoning using pesticide, for example, may be missed or misclassified as unintentional self-poisoning and can therefore be under-reported *(18)*. Similarly, some intentional self-harm may be masked as road traffic injury deaths. Additional information should be obtained to minimize the registration of deaths as "undetermined" wherever possible, as they may hide suicides and thus restrict surveillance and intervention *(18)*.

Fig. 2. *Distinguishing natural, unintentional, suicide and homicide deaths*



As described in section 2, deaths due to suicide often require medico-legal investigation of the circumstances of the death. Some of these additional sources of evidence may include forensic autopsy and police or coronial enquiries involving investigations of evidence at the scene and of the deceased's history, including substance use. Firearms and explosives trace reports, and Internet search history or social media use may also be investigated *(19)*. In settings with limited resources, certifiers may have to rely on basic information such as:

- the method and circumstances of death that suggest the death was self-inflicted rather than unintentional, natural or due to homicide;
- a medical history review to identify recent discharge from a mental health service, diagnosis of a mental health condition or severe chronic/terminal illness, recent hospitalization for self-harm, and consumption of medications, alcohol or drugs prior to death;
- the presence of written communication indicating intention to die (e.g. a suicide note or other written evidence which can be handwritten or electronically stored on computers or smartphones, or published on social media); and/or
- verbal communication indicating the intention to die.

Although factors related to suicide are likely to differ according to cultural, economic and social contexts, other information that may be collected to help determine the likelihood that the death was due to suicide include:

- evidence of stressful life events (e.g. loss of a loved one, relationship breakdown);
- evidence of considerable emotional distress at the time of death;
- evidence of financial problems (e.g. debt, job loss, business downturn);
- reports of familial disputes, domestic violence;
- substance use;
- a history of self-harm.

None of these factors are definitive evidence of suicide, but they may aid in determining the cause and manner of death. A more comprehensive list of items to help determine the likelihood that the death was due to suicide is provided in Annex 1. The annex is not designed to be used as a checklist, but it may help police officers, medical certifiers, and other professionals involved in the investigation of unnatural deaths to reach a decision more consistently regarding the cause and manner of death across a given jurisdiction. Further detail on external examination of the body and gathering of information from household members of the deceased for determination of the cause of death is provided in *WHO Recommendations for conducting an external inspection of a body and filling in the Medical Certificate of Cause of Death (3)*.

3.2 Certifying a death due to suicide using the WHO international form of the Medical Certificate of Cause of Death

The ICD's mechanisms for standardizing the procedures relating to causes of death include an international form of the Medical Certificate of Cause of Death (MCCD) (Fig. 3) *(20)*. The MCCD should not be confused with the death certificate, which is the legal document that families of the decedent receive once a death has been registered in the civil registration system *(3)*. Although not all countries are using the MCCD, and some may be using an older version of the MCCD,[7] for the purposes of this booklet, guidance is provided for completion of the latest WHO 2016 MCCD (Fig. 3). In countries or jurisdictions where there is no standard cause-of-death form or where a suicide case registry is in the course of being established, a suicide case registration form (such as the example in Annex 2) may be adapted according to the local context.

Once the necessary information about the deceased has been collected, the certifier can then complete the MCCD. For deaths due to suicide, this will often be a medical doctor. However, depending on the legal framework, a forensic doctor, police officer, coroner or even multiple authorities may be involved in completion of the form. A properly completed MCCD clarifies both the cause (the injury or

[7] As provided in an earlier revision of the ICD, for instance.

disease leading to death) and the manner (how that injury or disease occurred) of death. Ideally, the certifier should have knowledge of the medical history of the deceased person and the circumstances of the death or be able to access this information for the certification process. Certifiers should use their best medical opinion alongside the information obtained through additional investigation to determine and report the cause and manner of death on the MCCD.

ICD defines the "underlying cause of death" as "(a) the disease or injury which initiated the train of morbid events leading directly to death, or (b) the circumstances of the accident or violence which produced the fatal injury" *(20)*. The medical data in Frame A of the MCCD are in two parts: Part 1 is intended to certify the sequence of events that led to the death, while Part 2 contains other conditions that contributed to the death (Fig. 3). In Part 1 of the MCCD, there are four rows on which the sequence of events leading to death are recorded. The direct or immediate cause of death is entered on the first line (1a). There must always be an entry on this line.

In the case of a death due to suicide, if only the underlying cause is reported and no further sequence of events is recorded, the specific injury or mechanism leading to the death (e.g. "intentional self-inflicted gunshot wound to head") should be reported on the first line 1a, with subsequent lines kept blank. Where two or more conditions are reported (e.g. "intra-cranial haemorrhage" and "catastrophic temporal lobe injury"), the sequence of events leading to death should be entered on a separate line, and the underlying cause of death (e.g. "intentional self-inflicted gunshot wound to head") should be stated on the lowest used line. While the term "suicide" may be included in Part 1 to indicate the intentional nature of death, certifiers should always specify the precise mechanism or injury in order to enhance the quality of mortality data. The word 'suicide' alone should not be recorded as the cause of death.

In addition to completing the medical data of Frame A, the certifier should complete the necessary components of Frame B which cover other medical data (Fig. 3).

In the case of suicide, important components to be completed in Frame B include the manner of death; whether an autopsy was requested and whether information from the autopsy was used for certification; the date of injury; and the place of occurrence of the external cause. Regarding the manner of death, the option to be selected for suicide is "Intentional self-harm". Previous versions of the MCCD did not include the section on manner of death which was added in the 2016 MCCD. All efforts should be made to complete the section on the manner of death and to avoid reporting "could not be determined" or "unknown". Such responses have an impact on subsequent ICD coding of the MCCD and, may result in under-reporting of deaths. "Undetermined" manner of death should be reserved only for cases where no information exists, or no definitive manner of death could be determined even after thorough investigation.

The section on manner of death also includes a text box for further description. In the case of intentional self-harm, for example, it allows for including information on suicidal intent (if available) and how and where the external cause occurred and, if applicable, for specifying the poisoning agent. It is advised that the commercial name or the chemical name of the poisoning agent is specified, and that the physical characteristics of the formulation (e.g. liquid, gas, solid, powder) and route of exposure (e.g. oral, inhalation) are described *(21)*. The commercial name of the product may be obtained from first responders and/or family members who may find the product at the scene. Means restriction is an important suicide prevention strategy *(1)*. For instance, by recording the exact type of pesticide used (e.g. brand name or active ingredient), problematic substances can be identified, and this can inform the implementation of product bans or stricter regulation *(22)*.

In some medico-legal death investigation systems (e.g. coroner or hybrid systems that include an inquest process) the certifying doctor may present information to the coroner who then completes the MCCD, or may submit the MCCD with only the causes of death reported in Frame A. Regardless of the country's legal framework and processes that may ascribe authority to multiple persons for the completion of the MCCD, it is critical to ensure that all sections are completed accurately with the best available detail. Medical abbreviations should be avoided and clear, legible documentation is essential to ensure the accurate subsequent coding by mortality coders. The MCCD should be submitted to the registration authority (e.g. civil registration or national statistics office) in a timely manner for registration of the death and coding of the causes of death.

Fig. 3. *WHO 2016 International form of the Medical Certificate of Cause of Death (MCCD)*

Administrative Data (can be further specified by country)			
Sex	☐ Female	☐ Male	☐ Unknown
Date of birth	D D M M Y Y Y Y	Date of death	D D M M Y Y Y Y

***Frame A*: Medical data: Part 1 and 2**			
1 Report disease or condition directly leading to death on line a		Cause of death	Time interval from onset to death
	a		
Report chain of events in due to order (if applicable)	b	Due to:	
	c	Due to:	
State the underlying cause on the lowest used line	d	Due to:	
2 Other significant conditions contributing to death (time intervals can be included in brackets after the condition)			

***Frame B*: Other medical data**			
Was **surgery** performed within the last 4 weeks?	☐ Yes	☐ No	☐ Unknown
If yes please specify date of surgery	D D M M Y Y Y Y		
If yes please specify reason for surgery (disease or condition)			
Was an autopsy requested?	☐ Yes	☐ No	☐ Unknown
If yes were the findings used in the certification?	☐ Yes	☐ No	☐ Unknown

Manner of death:		
☐ Disease	☐ Assault	☐ Could not be determined
☐ Accident	☐ Legal intervention	☐ Pending investigation
☐ Intentional self harm	☐ War	☐ Unknown
If external cause or poisoning:	Date of injury	D D M M Y Y Y Y
Please describe how external cause occurred (If poisoning please specify poisoning agent)		

Place of occurrence of the external cause:			
☐ At home	☐ Residential institution	☐ School, other institution, public administrative area	☐ Sports and athletics area
☐ Street and highway	☐ Trade and service area	☐ Industrial and construction area	☐ Farm
☐ Other place (please specify):			☐ Unknown

Fetal or infant Death			
Multiple pregnancy	☐ Yes	☐ No	☐ Unknown
Stillborn?	☐ Yes	☐ No	☐ Unknown
If death within 24h specify number of hours survived		Birth weight (in grams)	
Number of completed weeks of pregnancy		Age of mother (years)	
If death was perinatal, please state conditions of mother that affected the fetus and newborn			
For women, was the deceased pregnant?	☐ Yes	☐ No	☐ Unknown
☐ At time of death	☐ Within 42 days before the death		
☐ Between 43 days up to 1 year before death	☐ Unknown		
Did the pregnancy contribute to the death?	☐ Yes	☐ No	☐ Unknown

3.3 Coding a death due to suicide using ICD

The ICD, developed and maintained by WHO, is an international standard diagnostic classification for mortality and morbidity that is used for epidemiological and health management purposes. The ICD is revised periodically and is currently in its eleventh revision (ICD-11), with many countries in various phases of implementing ICD-11.

The ICD includes coding rules and principles for determination of the underlying cause of death - "the disease or injury which initiated the train of morbid events leading directly to death, or the circumstances of the accident or violence which produced the fatal injury".[8] Once the MCCD is completed and is submitted to registration authority, the information contained in the form is typically translated into ICD codes[9] by a trained mortality coder, and in some settings, with the support of an automated tool (e.g. DORIS),[10] for selection of the underlying cause of death using the ICD coding rules. The rules and principles facilitate the determination of the underlying cause of death that is considered most informative from a public health perspective as it can inform policy and programmes in order to prevent premature mortality.

It is the responsibility of the mortality coder to assign accurately the correct ICD-10 or ICD-11 code to the injury and external cause or other conditions reported on the MCCD, and to apply the rules and principles systematically to identify correctly the underlying cause of death. Coders are advised (where possible) to communicate directly with the certifying doctor when clarification is needed, especially for complex cases or if terminology is ambiguous.

Unlike previous revisions of the ICD, ICD-11 makes provision for additional "postcoordination" to describe further important details relevant to the event.[11] This means that extension codes should be added to specify the object, pesticide or substance used in the injury or poisoning, the place of occurrence, the additional role of alcohol or drugs, and further aspects of intentional self-harm such as proximal risk factors (e.g. income-related problems, previous suicide attempt, presence of the intention to die). Including these postcoordination extension codes is valuable for suicide prevention because it enables the identification of relevant risk factors and specific methods of suicide (such as a particular pesticide or firearm) that may be targeted through intervention. General guidance on transitioning to ICD-11 is provided on the ICD-11 homepage. A detailed explanation of certification and coding using ICD-10 and ICD-11, alongside training materials, can also be found on the WHO ICD-11[12] and ICD-10[13] websites.

[8] See link in Annex 6 to WHO webpage on Reporting cause of death.

[9] For intentional self-harm codes see Annex 3 for ICD-11 and Annex 4 for ICD-10 codes.

[10] The Digital Open Rule Integrated cause of death Selection (DORIS) tool helps in selection of the underlying cause of death following the mortality rules of the ICD. For details, see the link in Annex 6.

[11] See link in Annex 6 for details on postcoordination.

[12] See link in Annex 6 for access to all aspects of ICD-11.

[13] See link in Annex 6 for access to all aspects of ICD-10.

3.4 Examples of certifying and coding a death due to suicide

Examples are illustrated below of how a death due to suicide can be recorded in Frame A (parts 1 and 2) and Frame B (other medical data and manner of death) of the WHO MCCD and subsequently coded using ICD-10 and ICD-11 codes. For ICD-11, postcoordination codes and injury stem codes are also included to provide further detail on the death.

Case example 1: A deceased male aged 64 years is found with a self-inflicted gunshot wound to the head made by a handgun. He was located at home and had a treatment history of depression and a history of adverse life events, including the recent death of his spouse and financial debt.

For certifiers completing cause-of-death form

Frame A: Medical data: Part 1 and Part 2				
1 Report disease or condition directly leading to death on line a Report chain of events in "due to" order (if applicable) State the underlying cause on the lowest used line			Cause of death	Time interval from onset to death
		a	Intra-cranial haemorrhage	1 hour
		b	Due to: Catastrophic temporal lobe injury	1 hour
		c	Due to: Intentional self-inflicted gunshot (handgun) wound to the head	1 hour
		d	Due to:	
2 Other significant conditions contributing to death (time intervals can be included in brackets after the condition)			Major depressive disorder, adverse life events – death of spouse and financial debt.	

Frame B: Other medical data			
Was **surgery** performed within the last 4 weeks?	☐ Yes	☒ No	☐ Unknown
If yes, please specify date of surgery			
If yes, please specify reason for surgery (disease or condition)			

Was an autopsy requested?	☒ Yes	☐ No	☐ Unknown
If yes, were the findings used in the certification?	☒ Yes	☐ No	☐ Unknown

Manner of death:		
☐ Disease	☐ Assault	☐ Could not be determined
☐ Accident	☐ Legal intervention	☐ Pending investigation
☒ Intentional self-harm	☐ War	☐ Unknown

If external cause or poisoning:	Date of injury:
Please describe how external cause occurred (If poisoning, please specify poisoning agent)	Intentional self-harm gunshot wound to the head using a handgun. History of financial problems, grief, depression.

Place of occurrence of the external cause:			
☒ At home	☐ Residential institution	☐ School, other institution, public administrative area	☐ Sports and athletics
☐ Street and highway	☐ Trade and service area	☐ Industrial and construction area	☐ Farm
☐ Other place (please specify):			☐ Unknown

For coders translating cause-of-death information into ICD codes

Using the information reported in Frame A (cause) and Frame B (manner and description of injury), the coder selects the following underlying cause of death:

ICD-10 code	X72 Intentional self-harm by handgun discharge
ICD-11 code	PC50.0 Intentional self-harm by projectile from a handgun *Postcoordination extension codes:* Object and location: &XE72J (Handgun) &XE266 (Home) Aspects of intentional self-harm: &XE6XD (Mental condition) &XE2FT (Unspecified manner of death of a relative, partner or friend) &XE3U9 (Income-related or financial problem) *Injury stem code:* NA07 (Intracranial injury)

Case example 2: An otherwise healthy female aged 21 years dies after being admitted to hospital following the ingestion of paraquat, a highly hazardous pesticide. She has no history of psychiatric, physical or financial problems contributing to the death. A recent dispute with a family member was noted following police investigation, alongside written communication expressing suicidal thoughts on her social media account. In addition, police investigation of her personal computer found an Internet search history of various suicide methods.

For certifiers completing cause-of-death form

Frame A: Medical data: Part 1 and Part 2				
1 Report disease or condition directly leading to death on line a Report chain of events in "due to" order (if applicable) State the underlying cause on the lowest used line			Cause of death	Time interval from onset to death
		a	Intentional paraquat self-poisoning by oral ingestion.	12 hours
		b	Due to:	
		c	Due to:	
		d	Due to:	
2 Other significant conditions contributing to death (time intervals can be included in brackets after the condition)			Interpersonal conflict with family member, written communication of suicidal thoughts, Internet search history of suicide methods.	

Frame B: Other medical data			
Was **surgery** performed within the last 4 weeks?	☐ Yes	☒ No	☐ Unknown
If yes, please specify date of surgery			
If yes, please specify reason for surgery (disease or condition)			
Was an autopsy requested?	☒ Yes	☐ No	☐ Unknown
If yes, were the findings used in the certification?	☒ Yes	☐ No	☐ Unknown

Manner of death:		
☐ Disease	☐ Assault	☐ Could not be determined
☐ Accident	☐ Legal intervention	☐ Pending investigation
☒ Intentional self-harm	☐ War	☐ Unknown

If external cause or poisoning:	Date of injury:
Please describe how external cause occurred (If poisoning, please specify poisoning agent)	Intentional ingestion of 100ml of paraquat (liquid) from bottle.

Place of occurrence of the external cause:			
☒ At home	☐ Residential institution	☐ School, other institution, public administrative area	☐ Sports and athletics area
☐ Street and highway	☐ Trade and service area	☐ Industrial and construction area	☐ Farm
☐ Other place (please specify):			☐ Unknown

For coders translating cause-of-death information into ICD codes

Using the information reported in Frame A (cause) and Frame B (manner and description of injury), the coder selects the following underlying cause of death:

ICD-10 code	X68 Intentional self-poisoning by and exposure to pesticides
ICD-11 code	PD03 Intentional self-harm by exposure to or harmful effects of pesticides. *Postcoordination extension codes:* Substance and location: &XM4SL8 (Paraquat) &XE266 (Home) Aspects of intentional self-harm: &XE17Z (Conflict in relationship with family member, partner, or friend) *Injury stem code:* NE61 (Harmful effects of or exposure to noxious substances, chiefly nonmedicinal as to source, not elsewhere classified)

Case example 3: A deceased male aged 24 years is found hanging from a ceiling beam in his home and upon autopsy a blood alcohol concentration (BAC) of 0.08g% was determined and compensated cirrhosis of the liver. Psychological autopsy including review of interviews with family revealed a history of alcohol abuse that began in his teenage years.

For certifiers completing cause-of-death form

Frame A: Medical data: Part 1 and Part 2				
1 Report disease or condition directly leading to death on line a Report chain of events in "due to" order (if applicable) State the underlying cause on the lowest used line			Cause of death	Time interval from onset to death
		a	Asphyxiation	1 hour
		b	Due to: Intentional self-harm by hanging	1 hour
		c	Due to:	
		d	Due to:	
2 Other significant conditions contributing to death (time intervals can be included in brackets after the condition)			Alcohol misuse and liver cirrhosis. Toxicology shows blood alcohol concentration of 0.08%.	

Frame B: Other medical data			
Was **surgery** performed within the last 4 weeks?	☐ Yes	☒ No	☐ Unknown
If yes, please specify date of surgery			
If yes, please specify reason for surgery (disease or condition)			
Was an autopsy requested?	☒ Yes	☐ No	☐ Unknown
If yes, were the findings used in the certification?	☒ Yes	☐ No	☐ Unknown

Manner of death:		
☐ Disease	☐ Assault	☐ Could not be determined
☐ Accident	☐ Legal intervention	☐ Pending investigation
☒ Intentional self-harm	☐ War	☐ Unknown

If external cause or poisoning:	Date of injury:
Please describe how external cause occurred (If poisoning, please specify poisoning agent)	Decedent hanged himself in his home from a ceiling beam. Toxicology shows blood alcohol concentration of 0.08% and interviews with family indicate problem alcohol use.

Place of occurrence of the external cause:			
☒ At home	☐ Residential institution	☐ School, other institution, public administrative area	☐ Sports and athletics area
☐ Street and highway	☐ Trade and service area	☐ Industrial and construction area	☐ Farm
☐ Other place (please specify):			☐ Unknown

For coders translating cause-of-death information into ICD codes

Using the information reported in Frame A (cause) and Frame B (manner and description of injury), the coder selects the following underlying cause of death:

ICD-10 code	X70 Intentional self-harm by hanging, strangulation and suffocation
ICD-11 code	PC71 Intentional self-harm by threat to breathing by hanging. *Postcoordination extension codes:* Object and location: &XE4H9 (Rope, string, or twine) &XE266 (Home) Alcohol use in injury: &XE1G3 (Alcohol use, suspicion, or evidence of alcohol use by the injured person) Aspects of intentional self-harm: &XE2Q7 (Substance abuse) *Injury stem code:* NF05 (Asphyxiation)

As shown in Fig. 2, suicide cases are sometimes hard to differentiate from deaths due to accident and assault. Below are examples relating to a death due to assault and accident.

Case example 4: A deceased male aged 46 years is found with a wound to the chest made by a knife. He was located in a quiet street. Police identified some criminal history. Forensic autopsy revealed the injury was not self-inflicted.

For certifiers completing cause-of-death form

Frame A: Medical data: Part 1 and Part 2				
1 Report disease or condition directly leading to death on line a Report chain of events in "due to" order (if applicable) State the underlying cause on the lowest used line			Cause of death	Time interval from onset to death
		a	Perforation of the heart	1 hour
		b	Due to: Knife stab wound to the chest	2 hours
		c	Due to:	
		d	Due to:	
2 Other significant conditions contributing to death (time intervals can be included in brackets after the condition)			Criminal history identified by police.	

Frame B: Other medical data			
Was **surgery** performed within the last 4 weeks?	☐ Yes	☒ No	☐ Unknown
If yes, please specify date of surgery			
If yes, please specify reason for surgery (disease or condition)			
Was an autopsy requested?	☒ Yes	☐ No	☐ Unknown
If yes, were the findings used in the certification?	☒ Yes	☐ No	☐ Unknown

Manner of death:		
☐ Disease	☒ Assault	☐ Could not be determined
☐ Accident	☐ Legal intervention	☐ Pending investigation
☐ Intentional self-harm	☐ War	☐ Unknown

If external cause or poisoning:	Date of injury:
Please describe how external cause occurred (If poisoning, please specify poisoning agent)	Decedent was stabbed by another person. Forensic autopsy indicated injury was not self-inflicted.

Place of occurrence of the external cause:			
☐ At home	☐ Residential institution	☐ School, other institution, public administrative area	☐ Sports and athletics area
☒ Street and highway	☐ Trade and service area	☐ Industrial and construction area	☐ Farm
☐ Other place (please specify):			☐ Unknown

For coders translating cause-of-death information into ICD codes

Using the information reported in Frame A (cause) and Frame B (manner and description of injury), the coder selects the following underlying cause of death.

ICD-10 code	X99 Assault by sharp object
ICD-11 code	PE30 Assault by being cut or pierced by knife, sword, or dagger *Postcoordination extension codes:* Object and location: &XE174 (Knife) &XE5NE (Public highway, street or road) *Injury stem code:* NB31.Y (Other specified injury of heart)

Case example 5: A female aged 78 years dies in the ambulance on the way to the hospital following a fall down a short flight of stairs at home witnessed by her daughters. There was a history of hypertension. Forensic autopsy identified fatal injuries to her head as a result of a fall.

For certifiers completing cause-of-death form

Frame A: Medical data: Part 1 and Part 2				
1 Report disease or condition directly leading to death on line a Report chain of events in "due to" order (if applicable) State the underlying cause on the lowest used line			Cause of death	Time interval from onset to death
		a	Fractured skull	1 hour
		b	Due to: Blunt force impact to head	1 hour
		c	Due to: Unintentional fall from stairs	1 hour
		d	Due to:	
2 Other significant conditions contributing to death (time intervals can be included in brackets after the condition)			Hypertension	

Frame B: Other medical data			
Was **surgery** performed within the last 4 weeks?	☐ Yes	☒ No	☐ Unknown
If yes, please specify date of surgery			
If yes, please specify reason for surgery (disease or condition)			
Was an autopsy requested?	☒ Yes	☐ No	☐ Unknown
If yes, were the findings used in the certification?	☒ Yes	☐ No	☐ Unknown

Manner of death:		
☐ Disease	☐ Assault	☐ Could not be determined
☒ Accident	☐ Legal intervention	☐ Pending investigation
☐ Intentional self-harm	☐ War	☐ Unknown

If external cause or poisoning:	Date of injury:
Please describe how external cause occurred (If poisoning, please specify poisoning agent)	Decedent lost her balance and fell down a flight of stairs less than one metre, as witnessed by her daughters.

Place of occurrence of the external cause:			
☒ At home	☐ Residential institution	☐ School, other institution, public administrative area	☐ Sports and athletics area
☐ Street and highway	☐ Trade and service area	☐ Industrial and construction area	☐ Farm
☐ Other place (please specify):			☐ Unknown

For coders translating cause-of-death information into ICD codes

Using the information reported in Frame A (cause) and Frame B (manner and description of injury), the coder selects the following underlying cause of death.

ICD-10 code	W10 Fall on and from stairs and steps
ICD-11 code	PA60 Unintentional fall on the same level or less than 1 metre *Postcoordination extension codes:* Object and location: &XE3HC (Stairs) &XE266 (Home) *Injury stem code:* NA02 (Fracture of skull or facial bones)

3.5 Essential data for suicide mortality statistics

At a minimum, the CRVS or any other registration system for suicide mortality should aim to capture basic information such as:

- sex;
- age;
- method of self-injury or self-poisoning – e.g. using ICD-11 or ICD-10 codes (Annex 3 and Annex 4);
- date of death;
- time of death;
- location where the person was found or carried out the suicide;
- place of death;
- place of residence.

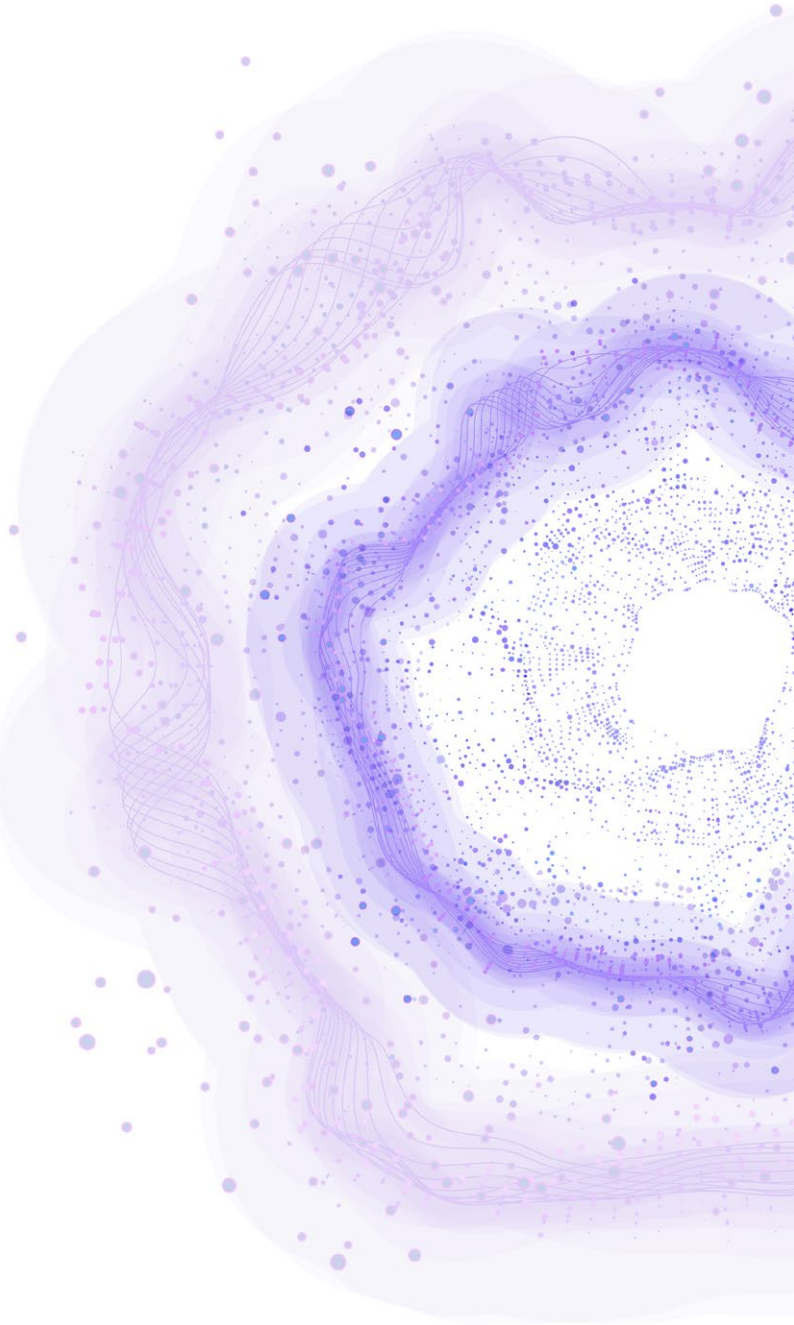
Many of these items would already be collected as part of routine death registration or police investigation. However, this information may be only partially available in some circumstances or jurisdictions. With regard to method(s) of suicide, it is very important that this information is as detailed as possible in order to be able to inform public health decisions and prevention activities.

There are also sociodemographic, clinical and contextual factors that are essential not only in helping to reconstruct the circumstances preceding the death but also in informing an understanding of risk and protective factors for suicide prevention. The importance of these items may vary by context and a suicide case registration form or MCCD form may be amended to include items such as:

- ethnicity;
- religion;
- gender identity;
- relationship status at the time of death;
- migration status (including refugees and asylum seekers);
- employment status at the time of death;
- occupation at the time of death;
- presence of written suicide communication – e.g. letter or social media post;
- verbal statement indicating an intention to die obtained from witnesses;
- consumption of alcohol or drugs immediately prior to death;
- history of suicide attempt(s);
- family/peer history of suicide;
- mental health conditions;
- recent discharge from a mental health service;
- chronic pain and/or physical illness;
- emotional distress at the time of death;
- grief or loss;
- interpersonal conflict or relationship breakdown;
- recent or lifetime history of physical, sexual and/or psychological abuse;

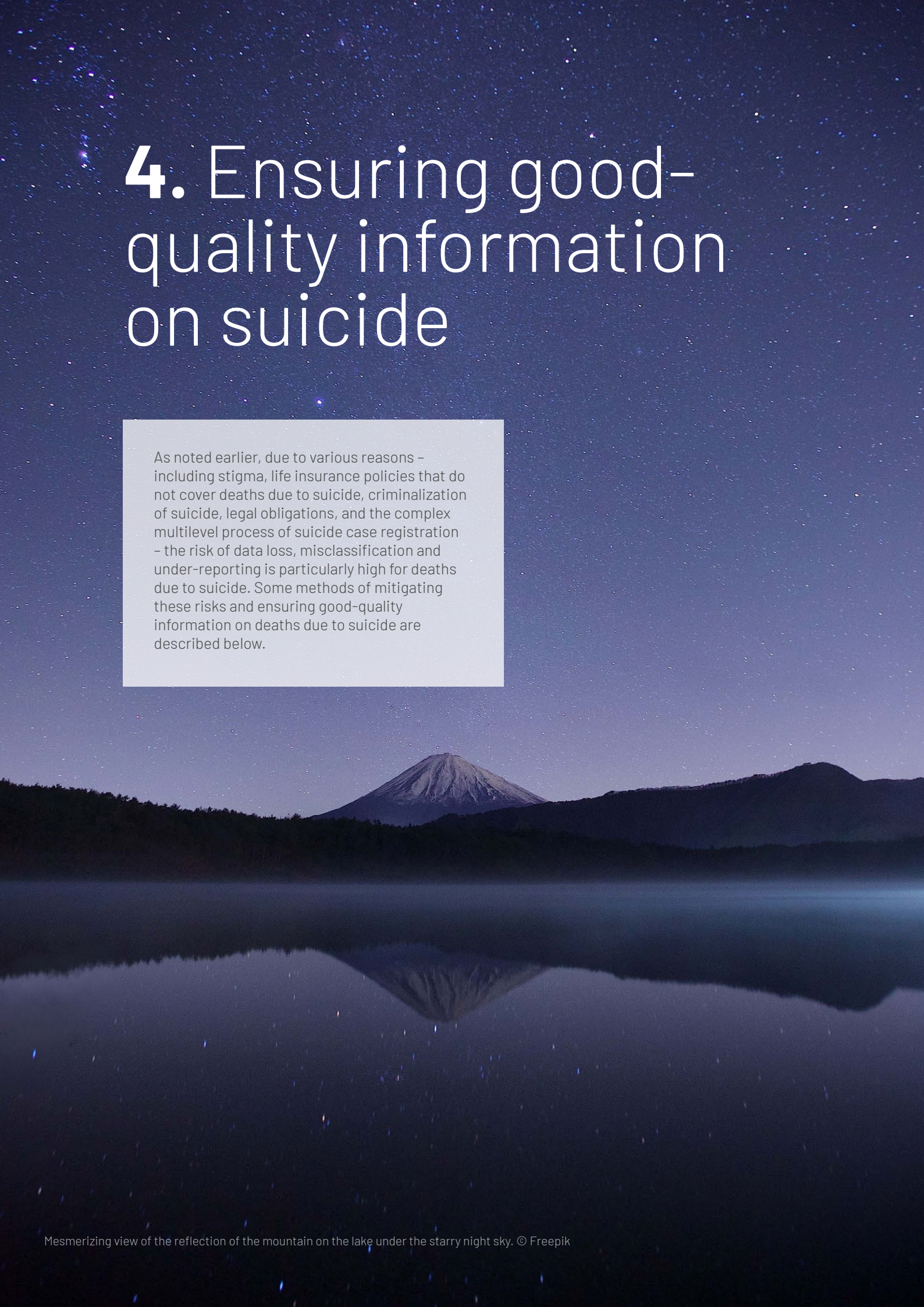
- financial problems (e.g. job loss, debt, business downturn);
- work or school-related issues (e.g. bullying, work/academic failure);
- experience of racism and discrimination;
- access to method (e.g. firearm stored in household, pesticide purchased from local vendor).

Sociodemographic and clinical information may be ascertained from medical records. However, detailed contextual information should be obtained from police investigation, key informants and possibly through psychological autopsy investigations (see Annex 5 for further information on psychological autopsy).



4. Ensuring good-quality information on suicide

As noted earlier, due to various reasons – including stigma, life insurance policies that do not cover deaths due to suicide, criminalization of suicide, legal obligations, and the complex multilevel process of suicide case registration – the risk of data loss, misclassification and under-reporting is particularly high for deaths due to suicide. Some methods of mitigating these risks and ensuring good-quality information on deaths due to suicide are described below.



Mesmerizing view of the reflection of the mountain on the lake under the starry night sky. © Freepik

4.1 Data management and data flow

Interagency collaboration and national regulations – across, for instance, the health sector (including health departments, hospitals), the medico-legal death investigation system, police, and statistical offices – may need to be established to allow seamless data flow. The workflow should clearly define pathways between the notification of death of a person, timely attendance of the person who has died by a medical certifier or other qualified certifier, completion of the MCCD form using all available information (i.e. external medical examination, police inquiry, and forensic autopsy if feasible), and submission of the cause-of-death information contained in the medical certificate to authorities in charge of civil registration and the national statistical office.

There is often a substantial delay between the occurrence of a death due to suicide and the availability of finalized and official statistics, which makes responsive suicide prevention activities, including suicide bereavement support, a challenge *(14)*. Timelines should be agreed between agencies. It could be that the death is registered with the civil registry office and the cause of death may be updated once finalized (e.g. following police/forensic investigation). For examples of interagency collaboration for suicide surveillance, see *(14)* and *(15)*.

4.2 Routine training and plausibility checks

It is important that those responsible for recording causes of death and coding suicide mortality information are routinely trained. This will help avoid errors and misclassifications. Ideally such training should begin during medical education and be continued in regular workshops and interactive in-service education programmes *(9)*.

A routine training programme led by clinical colleagues should be established to ensure that medical certifiers understand both the purpose of completing the MCCD form accurately and the form's importance for developing public health policies and preventing similar deaths. Staff who code the reported cause of death also need to be trained in how to apply the standard set of ICD rules for selection of the single underlying cause of death. International training materials are available on the WHO ICD-11[14] and ICD-10[15] websites. Establishing a system of supervision and oversight for both certifiers and coders can further support adherence to standards and improve the accuracy of cause-of-death documentation.

Plausibility checks on the distribution of data by sex, age and method, and feedback of the statistics (e.g. a summary statistical report) to medical certifiers should also be conducted routinely. This will help ensure that errors in the data can be promptly identified and corrected. Such reports could feature the level of missing data (e.g. the percentage of deaths due to suicide without a method specified).

[14] For information on the ICD-11 training package, see the relevant link in Annex 6.

[15] For information on the ICD-10 Interactive Self Learning Tool, see the relevant link in Annex 6.

WHO's online tool Analysing Mortality and Causes of Death 3 (ANACoD3)[16] is designed to monitor quality and trends using cause-of-death data. It highlights potential inconsistencies and errors in the data and estimates the completeness of reporting. In addition, WHO's CoDEdit tool[17] is applied at the data compilation stage to identify, for instance, possible misuse of codes and to flag basic errors in coding. Feedback of statistics on causes of death may also increase the motivation to invest time in accurate and complete reporting. Through refresher training, quality can be further maintained and even enhanced. In jurisdictions where staff retention and rotation can be an issue, train-the-trainer programmes on best practice are viable options to enhance standardization. A quick reference list of WHO resources, training material, and online tools related to the surveillance of mortality, including suicide, is provided in Annex 6.

4.3 Staff care

Mechanisms need to be in place to ensure the ongoing well-being of first-responders and those responsible for suicide death investigation, certification, data management and data use, including researchers *(23, 24)*. The well-being of staff is critical as it affects their ability to function in their role and to be able to support affected populations *(25)*. It is recommended that there should be adequate opportunity to debrief, receive supervision and ongoing training, receive recognition for work, and to ensure access to workplace-based psychosocial and mental health supports *(9, 25)*.

[16] For more details on ANACoD3, see Analysing Mortality and Causes of Death (ANACoD3) via the relevant link in Annex 6.

[17] For more details on CoDEdit, see Implementing basic checks on cause-of-death data via the relevant link in Annex 6.

5. Use of data collected on suicide mortality

Suicide mortality data can be of interest to a range of stakeholders including:

- government officials;
- policy-makers;
- researchers and scholars;
- health professionals;
- media professionals;
- community groups;
- national mortality registers;
- international organizations; and
- people with lived experience of suicide.

Providing suicide mortality data to policy-makers, researchers and health professionals allows for greater exposure of the scale of suicide and can be a way to raise awareness, initiate research, develop prevention campaigns, and monitor the effectiveness of suicide prevention and intervention strategies. The data can serve to inform stakeholders about specific trends in suicide, describing priorities in terms of (emerging) high-risk groups, and drawing attention to frequently used and emerging methods of suicide.

Bokeh background color disco. © Freepik.

5.1 Analysis and dissemination of suicide mortality statistics

A detailed explanation of how to perform statistical analysis and reporting is provided in WHO's *Practice manual for establishing and maintaining surveillance systems for suicide attempts and self-harm (26)* which can be similarly applied to suicide mortality data. Readers are advised to consult this manual for further instructions on data analysis and reporting, and on maintaining confidentiality. Further support in analysis can be found in the WHO ANACoD3[18] online tool. This tool automatically tabulates data and presents basic mortality measures in tables and figures. The tool allows for the analyses of cause-of-death data coded in ICD-10 as well as ICD-11 formats.

Data should be routinely analysed (e.g. annually). Reports on numbers, trends and relevant comparisons (e.g. by sex, age and method) should be presented. Numbers should be aggregated and categories of data with low numbers (e.g. fewer than five cases) should not be reported in order to protect confidentiality. To allow for comparison across jurisdictions, deaths by suicide should be reported as rates per 100 000 population. It is advised that statistics on suicide are reported responsibly, with careful attention paid to protecting privacy and to the language and style of reporting. Guidance on responsible reporting of suicide and self-harm can be found in WHO's *Preventing suicide: a resource for media professionals, update 2023 (27)*.

Data should be made available to relevant ministries and, if possible, the general public to inform decision-making and policy action. Data can be disseminated in a variety of ways, including through annual reports, evidence summaries, infographics and peer-reviewed publications.

The following outline may be used and can be adapted locally when generating a report on the statistical findings of data collected on suicide.

1. Executive summary
2. Methodology
    a. Definitions and terminology
    b. Data sources and recording
    c. Data coverage
    d. Analytical methods
3. Suicide statistics
    a. Overall and variation by age and sex
    b. Variation by method
    c. Variation by key sociodemographic characteristics (e.g. ethnicity/ religion, educational level, occupation)
    d. Variation by geographical region
    e. Variation by month
    f. Prevalence of other clinical/social factors (e.g. psychiatric diagnosis, domestic violence, economic distress, chronic illness).

[18] To upload data to ANACoD3 or to use a sample dataset, follow the link titled Welcome to ANACoD3, as listed in Annex 6.

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Annexes

Annex 1. Examples of information items to assist in determining a death due to suicide

Evidence of suicidal intent[1]

Direct

- Written expression of intent to die/end own life or of conflicting feelings ("ambivalence") about living or dying ("suicide note", including via letter, text, email, social media, video, etc).
- Verbal expression of intent to die/end own life or of conflicting feelings ("ambivalence") about living or dying.

Indirect (circumstantial)

Method

- Use of method which the deceased would be likely to consider highly lethal (e.g. hanging, gunshot, ligature, jumping from extreme height).
- Use of method which the deceased would be likely to consider lethal, with evidence of "full completion" of the method (e.g. taking all of the tablets contained in a packet of tablets or the entire contents of a bottle of medication).

Preparations

- Preparations for death which are inappropriate or unexpected in the context of the deceased's life.
- Effort to procure or learn about the means of death or to rehearse fatal behaviour.
- Recent accessing of suicide-promoting websites or social media.
- Active precautions to avoid rescue or genuine belief that adequate precautions had been taken to avoid being found, even if objectively not robust (e.g. returning to vacant property believing no-one would look there).
- Sorting out life affairs (e.g. giving away prized possessions or getting will in order).

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Psychological issues

- Expression of farewell or of the desire to die, or an acknowledgement of impending death.
- Expression of unbearable emotional or physical pain/distress.
- Expression of extreme hopelessness or other negative emotion regarding the future.
- Expression of inevitability of suicide or powerlessness to counter suicidal thoughts and feelings.
- Expression of perception of a failed struggle or feeling of powerlessness ("defeat").
- Expression of desire to hide or escape brought about by feeling that one's social status or public image has been threatened ("humiliation"/"shame").
- Expression of urgency to escape from an unbearable situation from which there is no perceived escape ("entrapment").
- Expression of unfulfilled desire for connection to other people ("thwarted belongingness").
- Expression of belief that one is a burden to society/others ("perceived burdensomeness").
- Expression of fearlessness about death.
- Unexplained improvement in mood (possibly signifying an enduring belief that suicide is a solution to their problems).
- Marked changes in behaviours – e.g. sleeping, eating, drinking (alcohol), drug-taking or other risk-taking behaviours.
- Acting unpredictably or out of character.
- Over-identification with, and dwelling on, thoughts ("ruminating") about a person who has died by suicide.

Capability for suicide

- Fearlessness about death.
- Tolerance of high level of physical pain.

Previous history of suicidal behaviour/intent and/or self-harm

- Previous history of self-harm (with or without intent to die) or suicide attempt (especially using highly lethal method).
- Previous expression of intent to die/kill oneself.

Current mental health conditions

- Recent discharge from mental health inpatient care/day care.
- Any current (treated or untreated) mental health condition (e.g. schizophrenia, depression, bipolar illness) or substance use disorder (alcohol/drugs).

Chronic and acute life events

- Long-term medical condition, especially if accompanied by chronic pain.
- Struggling to cope with recent stressful life event(s) – especially involving significant loss (e.g. death, particularly by suicide) of a loved one or rejection which led to marked change in self-esteem/social status).
- Interpersonal conflict within household or workplace/school setting.
- Recent or lifetime experience of abuse or violence.
- Experience of discrimination.
- Unemployment, recent job loss, business downturn, debt, or other financial stress.

Annex 2. Example suicide case registration form

1.1 Date of case registration (Day / Month / Year): _ _/_ _ / _ _ _	Not applicable	Unknown
1.2 Sample area (if applicable): _	888	999
2.1 Case name (if authorized): _	888	999
2.2 Case identification number (assigned): _ _ _ _ _ _ _ _ _	888	999
2.3 Residential address: _	888	999
2.4 Sex: _Male _Female _Intersex	888	999
2.5 Date of birth (Day / Month / Year): _ _ /_ _ / _ _ _ _	888	999
2.5.1 Age (in years) _ _ _ _ _ _	888	999
2.6 Date found (Day / Month / Year): _ _/_ _ / _ _ _ _	888	999
2.7 Day of the week found: _	888	999
2.8 Time found (Hour / Minute): _ _ / _ _	888	999
2.9 Date of death (Day / Month / Year): _ _/_ _ / _ _ _ _	888	999
2.10 Day of the week when died: _	888	999
2.11 Time of death (Hour / Minute): _ _ / _ _	888	999

2.12 Location found (Adapt to local context): _Home _Railway _..... _Other	888	999
2.12.1 Description of the location: _	888	999
3.1 Suicide method used: _	888	999
3.1.1 Description of suicide method (if intentional self-poisoning, describe the common name/commercial name of the substance or pesticide, physical characteristics of the formulation [liquid, gas, solid, powder], and route of exposure [e.g. oral, inhalation]): _	888	999
3.2 If more than one method was used, please list: _	888	999
3.2.1 Description of other suicide methods (if intentional self-poisoning, describe the common name/commercial name of substance/pesticide, physical characteristics of the formulation [liquid, gas, solid, powder] and route of exposure [e.g. oral, inhalation]): _	888	999
4.1 Is there written evidence of intent to die? _No _Yes	888	999
4.1.1 Description of evidence: _	888	999
4.2 Did the deceased make a verbal statement indicating an intention to die? _No _Yes	888	999
4.2.1 Description of evidence: _	888	999

Annex 3. ICD-11 codes for intentional self-harm[1]

Intentional self-harm by transport injury event

PB80	Land transport road traffic injury event
PB81	Land transport off-road nontraffic injury event
PB82	Land transport injury event unknown whether traffic or nontraffic
PB83	Railway transport injury event
PB90–PB9Z	Water transport injury event
PC00–PC0Z	Air or space transport injury event
PC2Y	Other specified transport injury event
PC2Z	Transport injury event, unspecified

Intentional self-harm by fall or jump

PC30	Fall or jump on same level or from less than 1 metre
PC31	Fall or jump from a height of 1 metre or more
PC3Y	Other specified fall or jump
PC3Z	Fall or jump, unspecified

Intentional self-harm by contact with person, animal or plant

PC40	Struck, kicked, or bumped by person
PC41	Struck, kicked, or bumped by animal
PC42	Stepped on or crushed by person
PC43	Stepped on or crushed by animal
PC44	Bitten by animal
PC45	Scratched or clawed by person
PC46	Scratched or clawed by animal
PC47	Stung or envenomated by animal
PC48	Contact with plant
PC4Y	Contact with person, animal or plant
PC4Z	Contact with person, animal or plant, type unspecified

[1] International Classification of Diseases Eleventh Revision (ICD-11). Geneva: World Health Organization; 2025 release. https://icd.who.int/en/.

Intentional self-harm by exposure to object, not elsewhere classified

PC50	Struck by projectile from firearm
PC51	Struck by moving object, not elsewhere classified
PC52	Striking against stationary object
PC53	Cut or pierced by sharp object
PC54	Struck by blunt object
PC55	Caught, crushed, jammed or pinched between objects
PC5Y	Contact with other specified type of weapon
PC5Z	Contact with weapon, type unspecified

Intentional self-harm by immersion, submersion or falling into water

PC60	Drowning or submersion while in body of water
PC61	Drowning or submersion following fall into body of water
PC62	Injury other than drowning while in body of water
PC63	Injury other than drowning following fall into body of water
PC6Z	Immersion, submersion or falling into water, unspecified

Intentional self-harm by threat to breathing

PC70	Threat to breathing by suffocation from object covering mouth or nose
PC71	Threat to breathing by hanging
PC72	Threat to breathing by strangulation
PC73	Threat to breathing by external compression of airways or chest
PC74	Inhalation or ingestion of gastric contents
PC75	Threat to breathing by inhalation or ingestion of liquids
PC76	Threat to breathing by inhalation or ingestion of food
PC77	Threat to breathing by inhalation or ingestion of other objects or materials
PC78	Threat to breathing from low oxygen environment
PC7Y	Threat to breathing
PC7Z	Threat to breathing, unspecified

Intentional self-harm by exposure to thermal mechanism

PC80	Exposure to controlled fire
PC81	Exposure to uncontrolled fire
PC82	Exposure to ignition or melting of material
PC83	Contact with hot object or liquid
PC84	Exposure to steam, hot vapour, air or gases
PC85	Exposure to excessive heat
PC86	Exposure to excessive cold
PC8Y	Exposure to other specified thermal mechanism
PC8Z	Exposure to unspecified thermal mechanism

Intentional self-harm by exposure to or harmful effects of substances

PC90	Exposure to or harmful effects of opioids or related analgesics
PC91	Exposure to or harmful effects of sedative hypnotic drugs or other CNS depressants
PC92	Exposure to or harmful effects of psychostimulants
PC93	Exposure to or harmful effects of cannabinoids or hallucinogens
PC94	Exposure to or harmful effects of analgesics, antipyretics or nonsteroidal anti-inflammatory drugs
PC95	Exposure to or harmful effects of antidepressants
PC96	Exposure to or harmful effects of antipsychotics
PC97	Exposure to or harmful effects of antiepileptics or antiparkinsonism drugs
PC98	Exposure to other or unspecified drug, medicament or biological substance
PC99	Exposure to or harmful effects of multiple drugs, medicaments or biological substances
PD00	Exposure to or harmful effects of alcohols
PD01	Exposure to or harmful effects of organic solvents
PD02	Exposure to or harmful effects of carbon monoxide
PD03	Exposure to or harmful effects of pesticides
PD04	Exposure to or harmful effects of corrosive substances
PD05	Exposure to or harmful effects of other or unspecified substances chiefly nonmedicinal as to source

Intentional self-harm by exposure to other mechanism

PD20	Foreign body in orifice
PD21	Exposure to electric current
PD22	Exposure to sunlight
PD23	Exposure to radiation
PD24	Exposure to high or low air pressure or changes in air pressure
PD25	Explosion or rupture of pressurised materials or object
PD26	Physical overexertion
PD27	Lack of food
PD28	Lack of water
PD29	Other specified privation
PD3Y	Other specified intentional self-harm
PD3Z	Intentional self-harm, unspecified

Annex 4. ICD-10 codes for intentional self-harm

X60	Intentional self-poisoning by and exposure to nonopioid analgesics, antipyretics and antirheumatics
X61	Intentional self-poisoning by and exposure to antiepileptic, sedative-hypnotic, antiparkinsonism and psychotropic drugs, not elsewhere classified
X62	Intentional self-poisoning by and exposure to narcotics and psychodysleptics [hallucinogens], not elsewhere classified
X63	Intentional self-poisoning by and exposure to other drugs acting on the autonomic nervous system
X64	Intentional self-poisoning by and exposure to other and unspecified drugs, medicaments and biological substances
X65	Intentional self-poisoning by and exposure to alcohol
X66	Intentional self-poisoning by and exposure to organic solvents and halogenated hydrocarbons and their vapours
X67	Intentional self-poisoning by and exposure to carbon monoxide and other gases and vapours
X68	Intentional self-poisoning by and exposure to pesticides
X69	Intentional self-poisoning by and exposure to other and unspecified chemicals and noxious substances
X70	Intentional self-harm by hanging, strangulation and suffocation
X71	Intentional self-harm by drowning and submersion
X72	Intentional self-harm by handgun discharge
X73	Intentional self-harm by rifle, shotgun and larger firearm discharge
X74	Intentional self-harm by other and unspecified firearm discharge
X75	Intentional self-harm by explosive material
X76	Intentional self-harm by smoke, fire and flames
X77	Intentional self-harm by steam, hot vapours and hot objects
X78	Intentional self-harm by sharp object
X79	Intentional self-harm by blunt object
X80	Intentional self-harm by jumping from a high place
X81	Intentional self-harm by jumping or lying before moving object
X82	Intentional self-harm by crashing of motor vehicle
X83	Intentional self-harm by other specified means
X84	Intentional self-harm by unspecified means
Y870	Sequelae of intentional self-harm

Annex 5. Psychological autopsy method

A psychological autopsy refers to in-depth investigations conducted by forensic investigators and/or researchers. A psychological autopsy (as distinct from a verbal autopsy) was initially developed to assist in situations where the cause of death was ambiguous or hard to determine as being an accident or suicide (Kõlves et al., 2021). A psychological autopsy typically consists of a review of all written case-related materials (police documents, medical records etc.) and interview(s) by trained interviewers with multiple informants (including family members) into the history, experiences, possible motivations and presence of other biopsychosocial factors (Kõlves et al., 2021). In view of the features of the psychological autopsy and the time it takes to be conducted, the primary purpose of psychological autopsy over the years has changed from being a tool to help assign the cause of death to being used for research purposes. More specifically, this method is commonly used in known cases of suicide to help determine the contributing factors and to enhance research on suicide prevention.

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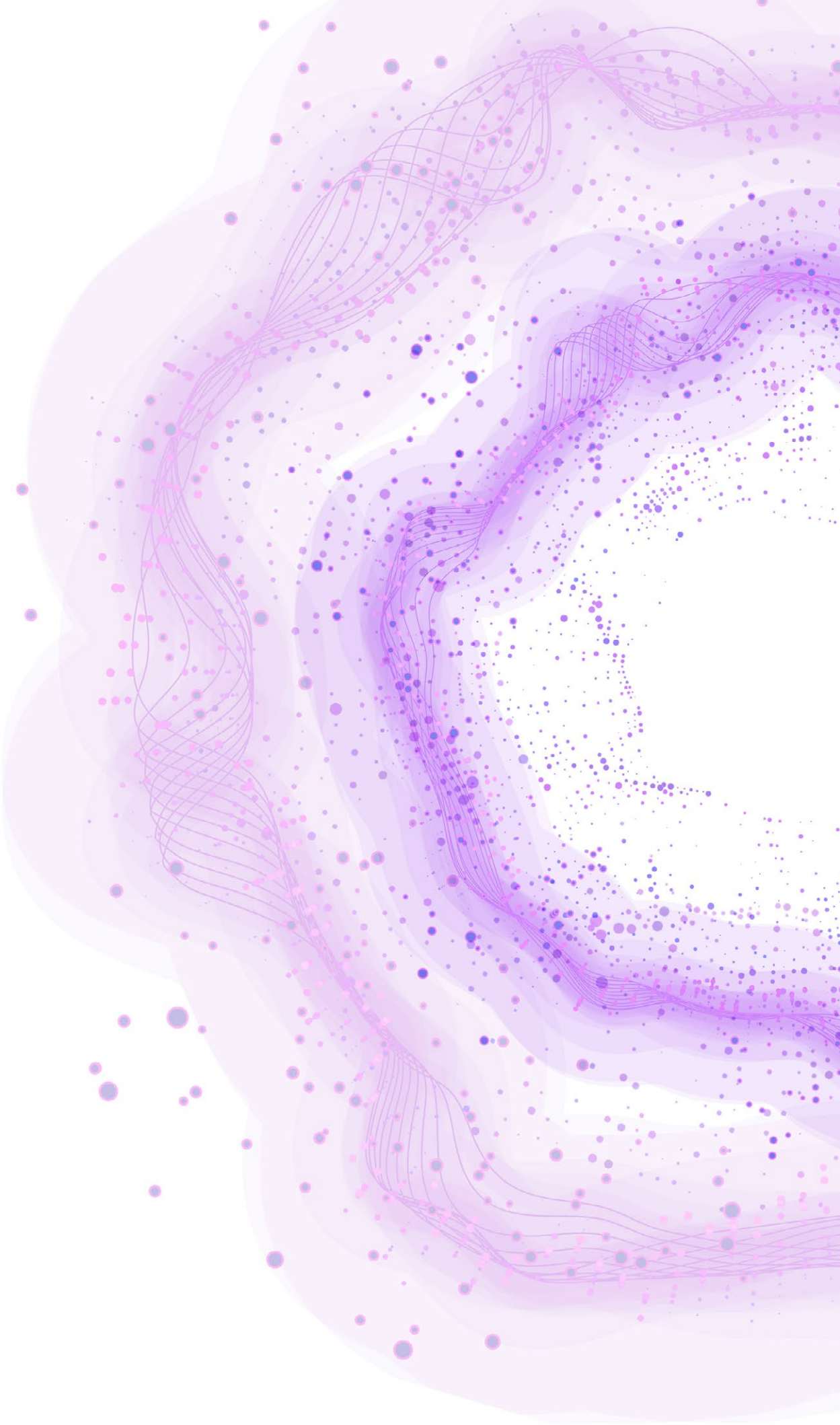
Annex 6: Quick reference list of related WHO resources on surveillance (non-exhaustive)

Technical documents

- WHO Recommendations for conducting an external inspection of a body and filling in the Medical Certificate of Cause of Death (2023): https://www.who.int/publications/m/item/who-recommendations-for-conducting-an-external-inspection-of-a-body-and-filling-in-the-medical-certificate-of-cause-of-death.
- Training manual for surveillance of suicide and self-harm in communities via key informants (2022): https://iris.who.int/handle/10665/365481.
- Practice manual for establishing and maintaining surveillance systems for suicide attempts and self-harm (2016): https://iris.who.int/handle/10665/208895.
- Strengthening civil registration and vital statistics for births, deaths and causes of death: resource kit (2013): https://iris.who.int/handle/10665/78917.
- Improving the quality and use of birth, death and cause-of-death information: guidance for a standards-based review of country practices (2010): https://iris.who.int/handle/10665/44274.

Online tools and training

- Suicide data: https://www.who.int/teams/mental-health-and-substance-use/data-research/suicide-data
- ICD-11: https://icd.who.int/en/
- ICD-11 Reference Guide: https://icdcdn.who.int/icd11referenceguide/en/html/index.html?sfvrsn=9ec05f86_1#/upload
- ICD-11 Postcoordination: https://icd.who.int/docs/codingtool/en/Postcoordination
- ICD-10: https://icd.who.int/browse10/2019/en.
- ICD-11 Training package: https://icdcdn.who.int/icd11training/index.html.
- ICD-10 Interactive self-learning tool: https://icdtraining.z16.web.core.windows.net/training/ICD10Training/.
- Digital Open Rule Integrated Cause of Death Selection (DORIS) tool: https://icd.who.int/doris/en.
- Analysing Mortality and Causes of Death 3 (ANACoD3) tool: https://www.who.int/standards/classifications/classification-of-diseases/services/analysing-mortality-levels-and-causes-of-death.
- CodEdit tool: https://www.who.int/standards/classifications/classification-of-diseases/services/codedit-tool.
- Verbal autopsy standards: https://www.who.int/standards/classifications/other-classifications/verbal-autopsy-standards-ascertaining-and-attributing-causes-of-death-tool.
- WHO STEPS survey: https://www.who.int/teams/noncommunicable-diseases/surveillance/systems-tools/steps.
- Reporting cause of death: https://www.who.int/standards/classifications/classification-of-diseases/cause-of-death.



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